CSA0502 – QUERY PROCESSING FOR DATA SCIENCE

Lab Solutions – Set 3

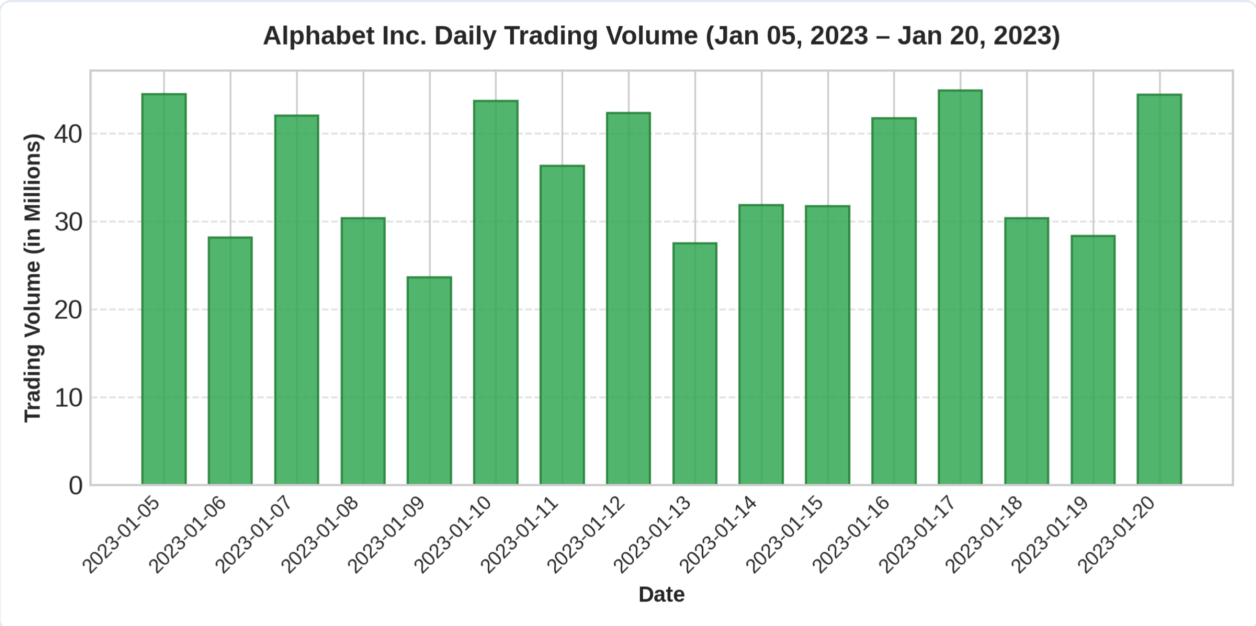
Question 5

> *Write a Pandas program to create a bar plot of the trading volume of Alphabet Inc. stock between two specific dates.*

Python Code:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Generate sample trading volume data for Alphabet Inc.
dates = pd.date_range(start='2023-01-05', end='2023-01-20')
np.random.seed(10)
volume = np.random.randint(20000000, 45000000, size=len(dates))

df_vol = pd.DataFrame({
    'Date': dates.strftime('%Y-%m-%d'),
    'Volume': volume
})

# Create bar plot of trading volume
plt.figure(figsize=(8, 4))
plt.bar(df_vol['Date'], df_vol['Volume'] / 1e6, color='#34a853', edgecolor='#1e7e34',
alpha=0.85, width=0.65)
plt.title('Alphabet Inc. Daily Trading Volume (Jan 05, 2023 – Jan 20, 2023)')
plt.xlabel('Date')
plt.ylabel('Trading Volume (in Millions)')
plt.xticks(rotation=45, ha='right')
plt.grid(axis='y', linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()
```

Output:



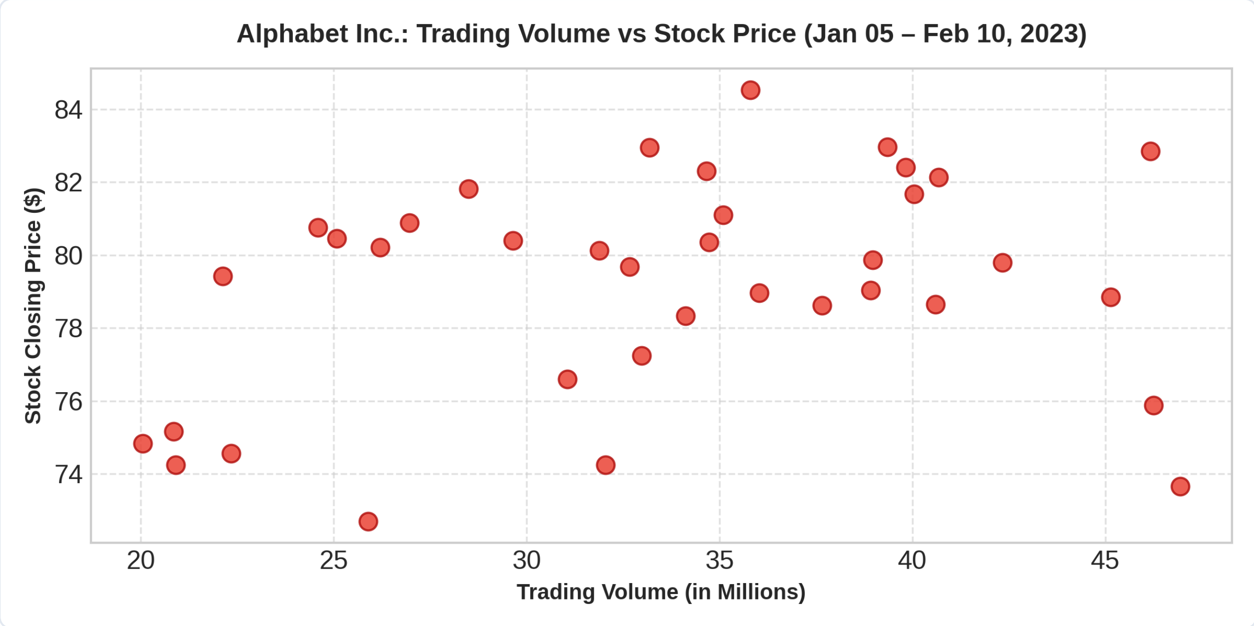
Question 6

> *Write a Pandas program to create a scatter plot of the trading volume/stock prices of Alphabet Inc. stock between two specific dates.*

Python Code:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Generate sample stock price and trading volume data
dates = pd.date_range(start='2023-01-01', end='2023-02-15')
np.random.seed(15)
close_prices = 85 + np.random.randn(len(dates)).cumsum() * 1.2
volume = np.random.randint(18000000, 50000000, size=len(dates))

df_scatter = pd.DataFrame({
    'Date': dates,
    'Close': close_prices,
    'Volume': volume
})

# Filter data between two specific dates
mask = (df_scatter['Date'] >= '2023-01-05') & (df_scatter['Date'] <= '2023-02-10')
df_filtered = df_scatter.loc[mask]

# Create scatter plot
plt.figure(figsize=(8, 4))
plt.scatter(df_filtered['Volume'] / 1e6, df_filtered['Close'], color='#ea4335',
alpha=0.85, edgecolors='#b31412', linewidths=1, s=65)
plt.title('Alphabet Inc.: Trading Volume vs Stock Price (Jan 05 – Feb 10, 2023)')
plt.xlabel('Trading Volume (in Millions)')
plt.ylabel('Stock Closing Price ($)')
plt.grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()
```

Output:


Alphabet Inc.: Trading Volume vs Stock Price (Jan 05 – Feb 10, 2023)
Stock Closing Price ($)
84
82
80
78
76
74
20
25
30
35
40
45
Trading Volume (in Millions)